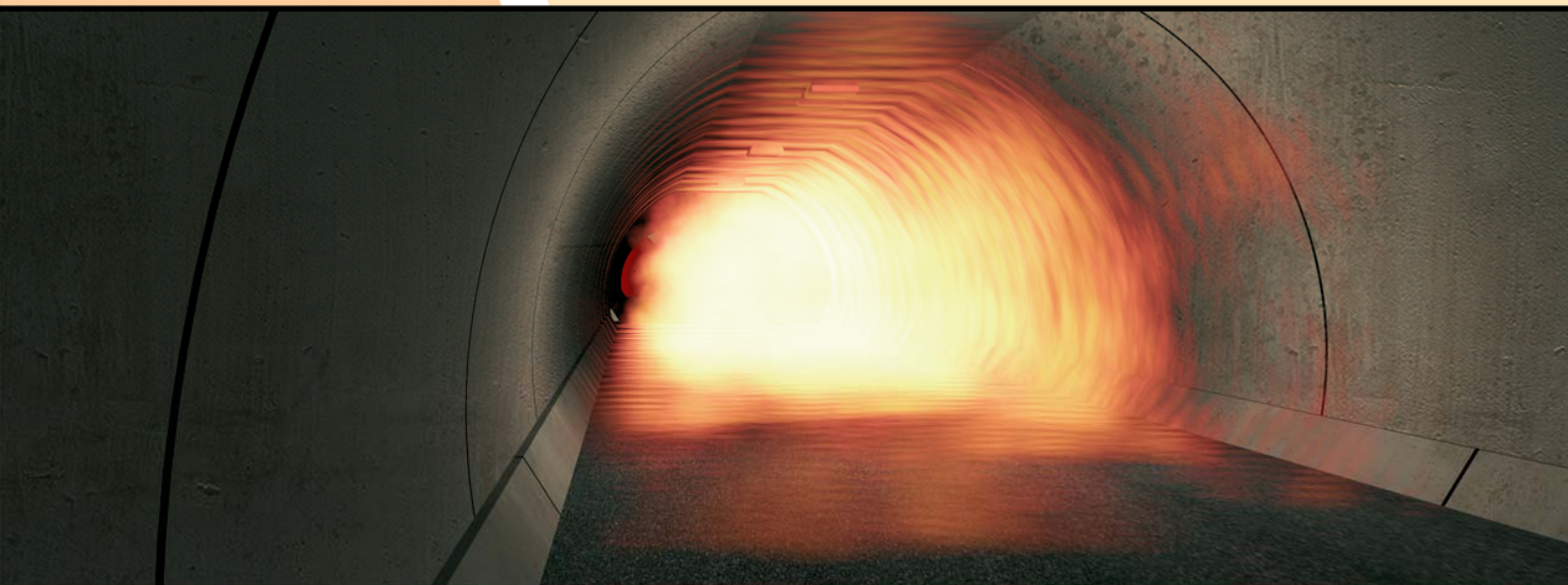


Fire In Tunnels

Thematic network



Technical report part 2

Fire Safe Design

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Thematic Network FIT – Fire in Tunnels

Technical Report – Part 2 Fire Safe Design

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Overview of the FIT reports

The Thematic Network FIT 'Fire in Tunnels' aims to establish and develop a European platform and optimise efforts on fire safety in tunnels. The Network's ambition is to develop a European consensus on fire safety for road, rail and metro tunnel infrastructures and enhance the exchange of up-to-date knowledge gained from current practice and ongoing European and national research projects.

The outcome of the FIT network is presented in 3 complementary formats:

- FIT website (www.etnfit.net)
- General report
- Technical Reports on
 - o Design fire scenarios;
 - o Fire safe design; and
 - o Fire response management

The **FIT website** (www.etnfit.net) contains the 6 consultable databases, the co-membership, the presentations of the International Symposium on Safe and Reliable Tunnels (Prague 2004) and the technical reports. The reports are available after registration as a corresponding member.

The **General report** presents the outcome of the FIT activities. After the introduction of the FIT Network, the **general approach to tunnel fire safety** is presented. This chapter can be considered as a strategic introduction to the consecutive safety aspects and the integrated approach to safety in tunnels. It introduces the highlights of the technical reports of the FIT network with the **executive summaries on design fire scenarios, fire safe design and fire response management**.

The **Technical reports** on the FIT workpackages presents the detailed reflexion and results of the network on the items in more than 450 pages state of the art research work. The reports are available from the FIT website after registration as a corresponding member.

Technical report Part 1 'Design fire scenarios' describes recommendations on design fire scenarios for road, rail and metro tunnels. Design fires to cover different relevant scenarios (e.g. design fires referring to the evacuation of people, design fires referring to ventilation purpose or design fires referring to the structural load) are presented and recommended.

In **Technical report Part 2 'Fire Safe Design'**, a compilation of relevant guidelines, regulations, standards or current best practices from European member states (and important tunnel countries like e.g. Japan and USA) is given. The analysis is focused on all fire safety elements regarding tunnels properly said and are classified according to the transport nature: road, rail and metro.

The occurrence of a fire in a tunnel provokes a need for response from the tunnel users, the operators and the emergency services. The **Technical report Part 3 'Fire response management'** presents the best practices which should be adopted by these different categories to ensure a high level of safety.

The **Technical reports** on the FIT workpackages presents the detailed reflexion and results of the network on the items in more than 450 pages state of the art research work. The reports are available from the FIT website after registration as a corresponding member.

Technical report – Part 1 ‘Design fire scenarios’

Rapporteur Alfred Haack, STUVA

The technical report of FIT Work Package 2 is devoted to design fire scenarios for road, rail and metro tunnels. It collects data from different countries (e.g. Germany, France, Italy, UK), international organisations (e.g. PIARC, ITA, UPTUN) as well as from the experiences in individual tunnels (e.g. Mont Blanc, Tauern, Nihonzaka, Caldecott, Pfänder). The report includes basic principles of design fires, tunnel fire statistics and impacts of fires and smoke in tunnels on people, equipment and structure. The data is analysed and different sets of data are compared to ascertain the degree of confidence attributed to the information. Recommendations are made within the text on specific issues when this was deemed appropriate and reliable.

Technical report – Part 2 ‘Fire Safe Design’

Rapporteur Bruno Brousse, CETU

Fire Safe Design – Road	Niels Peter Hoj, COWI
Fire Safe Design – Rail	Giorgio Micolotti, RFI
Fire Safe Design – Metro	Daniel GABAY, Arnoud Marchais, RATP

The FIT Workpackage ‘Compilation of guidelines for fire safe design’ presents the compilation of relevant guidelines, regulations, standards or current best practices from European member states, including reference documents from important tunnel countries like e.g. USA and Japan, or from European or international organisations, e.g. PIARC and UN/ECE. The report is classified according to the transport nature in three similar main sections: road, rail and metro tunnels. The three sections in the report presents the collected guidelines and regulations, their analytical abstract and table of content. About 50 safety measures are presented and compared related to structural measures (19), safety equipment (36) and structure and equipment with response to fire (3). For each type of measure the impact on safety is presented with a synthesis and a detailed comparison of the comprehensive list of safety measures.

Technical report – Part 3 ‘Fire Response Management’

Rapporteur Norman Rhodes, Mott MacDonald

The objective of the FIT Work Package 4 ‘Best practise for Fire Response Management’ is the definition of best practices for tunnel authorities and fire emergency services on prevention and training, accident management and fire emergency operations. The occurrence of a fire in a tunnel provokes a need for response from the tunnel users, the operators and the emergency services. The technical systems which are installed in many tunnels are described in Chapter 2. These systems contribute to the possible levels of safety that can be achieved and are mentioned later in relation to response planning. The viewpoint of the fire brigade is then presented in Chapter 3 in order to establish the context of fire response management. Best practices for Road, Rail and Metro tunnels then follow in Chapter 4, 5 and 6 respectively. They are presented according to the conceptual phases “before”, “during” and “after” a fire, taking into account the different involved parties (users, operators and emergency services).

Structure of report

Technical report Part 2 – Fire Safe Design

1	INTRODUCTION	20
1.1	Goal-approach method	
1.2	Scope of the compilation	
1.3	Comprehensive list of safety measures	
2	COMMENTS OF ROAD / RAIL / METRO COMPARISON	25
2.1	Investigation to harmonise guidelines for fire safe design	
2.2	General data on tunnels	
2.3	Tunnels in safety in view of the general operation for the three transport modes	
2.4	Traffic nature and potential fires	
2.5	Action towards fires	
2.6	Comparative synthesis table	
3	CONCLUSIONS ON THE COMPILATION OF GUIDELINES FOR FIRE SAFE DESIGN FOR ROAD, RAIL AND METRO TUNNELS	43
3.1	Main features identified by the guideline compilation	
3.2	More specifically for road tunnels	
3.3	More specifically for rail tunnels	
3.4	More specifically for metros	
3.5	Future work on fire safe design	

Technical Report Part 2: Fire Safe Design – Road Tunnels

1	LIST OF COLLECTED GUIDELINES	58
1.1	Table of references (national guidelines)	
1.2	Table of references (other reference documents)	
1.3	Analytical summaries (national guidelines)	
2	COMPREHENSIVE LIST OF SAFETY MEASURES	73
2.1	Structural measures relevant to safety	
2.2	Safety equipment	
2.3	Structure & equipment, response to fire	
3	MATRIX OF GUIDELINES CONTENTS	75
4	DETAILED COMPARISON	77
4.1	Structural measures relevant to safety	
4.2	Safety equipment	
4.3	Structure & equipment, response to fire	
4.4	Tunnel Classification	
5	APPENDIX 1: TABLES OF CONTENTS OF NATIONAL GUIDELINES TRANSLATED INTO ENGLISH	121
6	APPENDIX 2: TABLES OF CONTENTS OF OTHER REFERENCE DOCUMENTS TRANSLATED INTO ENGLISH	143

Technical Report Part 2: Fire Safe Design – Rail Tunnels

1	LIST OF COLLECTED GUIDELINES.....	160
1.1	Table of references (national# Guidelines)	
1.2	Table of references (other reference documents)	
1.3	Analytical summaries (national# guidelines)	
2	COMPREHENSIVE LIST OF SAFETY MEASURES	185
2.1	General design characteristics	
2.2	Structural measures relevant to safety	
2.3	Safety equipment	
2.4	Structure & equipment response to fire	
2.5	Emergency management	
3	MATRIX OF GUIDELINES CONTENTS	187
4	DETAILED COMPARISON.....	36
4.1	General Design Characteristics	
4.2	Structural measures relevant to safety	
4.3	Safety equipment	
4.4	Structural & equipment response to fire	
4.5	Emergency management	
4.5	Organisational measures	
5	APPENDIX 1: TABLES OF CONTENTS OF NATIONAL GUIDELINES (TRANSLATED INTO ENGLISH)	78
6	APPENDIX 2: TABLES OF CONTENTS OF OTHER REFERENCE DOCUMENTS (TRANSLATED INTO ENGLISH)	114

Technical Report Part 2: Fire Safe Design – Metro Tunnels

1	LIST OF COLLECTED GUIDELINES.....	286
1.1	Table of references for tunnels (national guidelines)	
1.2	Table of references for stations (national guidelines)	
1.3	Analytical summaries (national guidelines)	
2	COMPREHENSIVE LIST OF SAFETY MEASURES	292
2.1	Structural measures relevant to safety	
2.2	Safety equipment	
2.3	Structure & equipments, response to fire	
3	MATRIX OF GUIDELINES CONTENTS	294
3.1	Structural measures relevant to safety	
3.2	Safety equipment	
3.3	Structure & equipment response to fire	
4	APPENDIX 1: TABLES OF CONTENTS OF NATIONAL GUIDELINES FOR TUNNELS AND STATIONS TRANSLATED INTO ENGLISH	318



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FIT Co-membership

The FIT partnership is strengthened with a co-membership (co-opted members and corresponding members) to receive ample feedback and input and obtain a larger forum for the dissemination of its outcome.

The objectives of the corresponding and co-opted membership is the following:

- provide a large platform for the FIT working items
- ensure European feedback and input via organizations active in 'fire in tunnels'
- ensure member-state support via national and regional representatives

Co-opted members are organisations invited to contribute to the FIT activities in a very intensive way. They have the same access level as FIT network members (working document, etc.). Co-opted members are bound by an agreement of collaboration and confidentiality. Seventeen organisation have been invited and agreed as FIT Co-opted members.

Corresponding members further enlarge the FIT Network. Corresponding members are these organisations and national representatives that are interested to follow closely the activities of FIT and registered themselves via the FIT website. They have a privileged access to the endorsed FIT working documents and the Consultable Databases on fire and tunnel. A FIT public working document is a draft document that is being prepared for final edition by the FIT network. It is made available for the FIT corresponding members for consultation, input and comment.

More than 1200 corresponding members have been registered on the FIT website www.etnfit.net (status March 2005).

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Technical Report – Part 2

Fire Safe Design

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Table of contents

Chapter 1 : Introduction	20
1.1 Goal-approach method	20
1.2 Scope of the compilation	21
1.3 Comprehensive list of safety measures	22
Chapter 2 : Comments of road / rail / metro comparison	25
2.1 Investigation to harmonise guidelines for fire safe design	25
2.2 General data on tunnels	30
2.3 Tunnels in safety in view of the general operation for the three transport modes	34
2.4 Traffic nature and potential fires	35
2.5 Action towards fires	38
2.6 Comparative synthesis table	41
Chapter 3 : Conclusions on the Compilation of guidelines for fire safe design for road, rail and metro tunnels	43
3.1 Main features identified by the guideline compilation	43
3.2 More specifically for road tunnels	44
3.3 More specifically for rail tunnels:	46
3.4 More specifically for metros	47
3.5 Future work on fire safe design	48

Chapter 1 : Introduction

1.1 Goal-approach method

With very few exceptions a tunnel is not a dangerous risk of fire in itself, because it is nearly always of mineral constitution (rock or concrete); the possible constructed sidewalls are selected as non-inflammable or hardly inflammable, the installed facilities do not present a heavy heat release rate, and the selected electric cables do not propagate fire.

The actual danger comes from outside the tunnel and from any mobile element penetrating into the tunnel.

Underground structures are built to enable all terrestrial transport modes to pass through:

- on a track: pedestrians or cyclists, even skiers in mountainous areas
- on a road: motor cycles, cars, buses, vans, small or large lorries
- on rail: passenger and freight trains, metros, tramways, funiculars
- on water channel: commercial or pleasure boats.

Furthermore the tunnels may pass under water, under urban areas and under mountains and hills.

The European Thematic Network FIT decided to examine the three transport modes used in Europe which use most largely tunnels: road vehicles, trains, and metros. The web site www.etnfit.net gives under the title “Regulations” the “compilation of guidelines for fire safe design” defined by Workpackage 3 on these three modes.

The objectives of working package 3 are the compilation of relevant guidelines, regulations, standards or current best practices from European member states. For road tunnels, we also introduced the reference to the recent European directive and also included relevant documents from important tunnel countries like e.g. USA and Japan, or from European or international organisations, e.g. PIARC and UN/ECE.

This compilation report is classified according to the transport nature in three main sections:

- Fire Safe Design – road tunnels
- Fire Safe Design – railway tunnels
- Fire Safe Design – metro tunnels

Each section includes four chapters. Chapter 1 presents:

- a table with the list of collected documents with the following information:
 - o the document title in its original language
 - o the reference code
 - o the publishing date
 - o the administrative value
 - o possible comments on application, especially on enforcement conditions
- the analytical abstract of national tunnel regulations in the various countries
- as Appendix, the table of contents, translated into English, of the analysed national documents.

Chapter 2 recalls the selected comprehensive list of safety measures. Chapter 3 gives a global presentation of a matrix of guidelines contents showing only those parameters which appear to be consistently dealt with national tunnel regulations. Chapter 4 is devoted to the detailed comparison of the comprehensive list of safety measures.

For reasons of a too heavy work volume the number of documents selected for this analysis was willingly restricted. The main countries (road and rail) or main cities (metro) were selected according to the available information and their interest. For a homogeneous reading of the document, relevant information has been recorded in a structure way using exactly the same list of safety measures for road, rail and metro as that given below in Chapter 1.3 Comprehensive list of safety measures.

The following elements are given for each nature of measures:

- The role of the measure: which is the objective aimed at? What is the impact on safety?
- A synthesis with comments: what can be deduced from the various national regulations?
- A comparison table: giving the detail of prescribed safety measures.

The conclusions on the global balance of the compilation will be given in Chapter Chapter 3 : (before the individual technical reports on road, rail and metro tunnels.

1.2 Scope of the compilation

The analysis of WP3 'Compilation of guidelines for fire safe design' is focused on all fire safety elements regarding tunnels properly said, thus excluding the intrinsic safety measures also planned in connected underground structures either existing due to their nature, e.g. metro stations, or likely to exist, such as railway stations or car parks or bus stations for road.

Metro, however, must be considered as a very special underground structure, showing very close intermediate access and safety premises due to the numerous stations distributed along the metro line. The analysis obviously considers this essential safety element.

For rail and metro the construction standards of the rolling stock concerning fire, also the fire safety facilities in trains are not included in the evaluation; but it should be kept in mind that, in most countries, they may reduce risks significantly in the case of underground urban travellers transport and metro. As a preamble this aspect is mentioned at the beginning of Technical report Part 2 - 'Fire Safe Design – metro tunnels. Regarding the communication means within the vehicles (radio or cellular phones) they also can have a predominant role in safety as described below.

Except regarding the fire behaviour of structures and facilities, the document analysis does not deal with the tunnel constructive aspect properly said, but essentially focuses on the safety measures peculiar to the tunnel to reduce the fire consequences. Such arrangements concern three specific fields:

- structural safety facilities
- safety equipment
- reaction/resistance to fire.

The preventive safety facilities, essentially based on the tunnel geometrical design and operating means and rules are described in the introductory part 'general approach to tunnel fire safety' and the Technical report Part 3 'Fire response management' respectively.

It should be noticed that limits between the prevention field and mitigation field sometimes overlap, because they can use the same equipment or human basis. Moreover some safety measures defined according to other objectives than a fire case, for instance current operating works, however may have a noteworthy favourable impact on fire risks (e.g. reduced accident risk due to restricted vehicle speed, traffic control to preserve smoothness, road police controls, etc.).

Lastly, either urged or not by the regulations of a given country, the possibility to define and optimise some safety measures on the basis of the integrated approach to safety in tunnel (or performance based design) using risk studies and fire engineering is mentioned in the report. Generally this type of approach can be considered only for major new structures, justifying the intervention of exceptional design teams and control and safety committees. Section 4 on the detailed analysis of safety measures obviously can be founded essentially on the prescriptive part of the governmental texts that the designer must then strictly observe.

1.3 Comprehensive list of safety measures

The comprehensive list of safety measures used for the technical comparison of guidelines is developed below.

1.3.1 Structural measures relevant to safety

- S1 Emergency passenger exit for users
 - S11 Parallel escape tube
 - S12 Emergency cross-passage
 - S13 Shelter
 - S14 Direct pedestrian emergency exit
- S2 Emergency access for rescue staff
 - S21 Separate emergency vehicle gallery access
 - S22 Cross passage vehicle access
 - S23 Emergency lane
 - S24 Direct pedestrian access (lateral, upstairs, shaft)
 - S25 Turning areas
 - S26 Firemen station at portals
- S3 Drainage of flammable liquids
 - S31 Inclination of tunnel axis
 - S32 Separate drainage systems
 - S33 Liquid sump
 - S34 Non porous surface course
- S4 Others

1.3.2 Safety equipment

- E1 Smoke control ventilation
 - E 11 Natural ventilation by shafts
 - E 12 Longitudinal
 - E 13 Transversal
 - E 14 Ventilation control sensors
 - Opacity
 - CO
 - NO_x
 - Anemometers
 - Counter pressure measurement at portals
- E2 Emergency exit and rescue access ventilation
- E3 Lighting measurement at portals
 - E31 Emergency tunnel lighting
 - E32 Marker light in tunnel
 - E33 Emergency exit and rescue access lighting
- E4 Signage (permanent/variable)
 - E41 Traffic signals outside the tunnel
 - E42 Traffic signals inside the tunnel
 - E43 Exit pedestrian signs
 - E44 Rescue pedestrian signs
- E5 Communication and alarm system
 - E51 Emergency telephone
 - E52 Alarm push button (manual fire alarm)
 - E53 Automatic alarm on equipments (exit doors, extinguisher, fire boxes)
 - E54 Automatic incident detection
 - E55 Fire/smoke detection (ventilation sensors or specific fire detection)
 - E56 Radio rebroadcast
 - tunnel users
 - emergency team
 - operator
 - E57 Loudspeakers (in tunnel, in shelters)
- E6 Traffic regulation - monitoring equipments
 - E61 Monitoring of traffic speed and intensity
 - E63 Close circuit television
 - E64 Remote control barriers
 - E66 Thermographic portal detectors (trucks)
- E7 Power supply
- E8 Fire suppression (fire fighting equipment)
 - E 81 First and fire fighting (extinguisher, hose-reels, etc ...)
 - E82 Fire fighting media
 - E84 Fixed fire suppression mitigation system (Sprinkler, Deluge)
- E9 Others

1.3.3 Structure & equipment, response to fire

R1 Reaction to fire

R2 Structure resistance to fire

R3 Equipment resistance to fire

- cables

- fans

Chapter 2 : Comments of road / rail / metro comparison

The safety problems are quite different, because each of these modes have very distinct and specific features regarding both the tunnel infrastructure and the nature of vehicles or rolling stock, and the operating rules. The accidentology level, the consequences of a vehicle fire in the tunnel, and the means to manage the fire response are not similar. The following tends – without the care of any exhaustiveness – to set up a first comparison of the problems laid by the fire safety in tunnel for these three modes of transport.

The causes of intentional fire are not considered.

The comments below propose a comparative synthesis of fire safety problems encountered in tunnels for road, rail and metro. It does not deal either with the fire behaviour aspects of structures and equipment or the drainage aspects.

The five topics dealt with successively are listed below and are concluded with a comparative synthesis table.

1. investigation to harmonize the guidelines for safe design
2. general data on tunnels
3. place of tunnels in safety of the general operation for the three transport modes
4. traffic nature and potential fires
5. action towards fires.

2.1 Investigation to harmonise guidelines for fire safe design

The dramatic fires which occurred in the road tunnels of Mont Blanc (France-Italy; 39 fatalities) and Tauern (Austria; 12 fatalities) in 1999 have caused a radical change of views on tunnel safety. This topic, which was previously reserved for specialists, became a real concern for the European public opinions, which triggered politicians to be involved. This concern was reinforced two years later by the fire in the Gotthard tunnel (Switzerland; 11 fatalities). Rail tunnels were also affected by fire catastrophes, such as in the Channel tunnel (UK-France; no fatality but very severe damage) in 1996, Kaprun funicular tunnel (Austria; 155 fatalities) in 2000 or Daegu metro (South Korea; 200 fatalities) in 2003.

Of course tunnel fire safety had been studied for a long time before these fires, so that important knowledge was available, as well as a number of recommendations and regulations. However these were considered insufficient, so that a number of new initiatives have been launched in individual countries and at the European and international levels. These include research works, networking activities and development of new regulations.

2.1.1 Regarding road

Situation before 1999

Even though public opinions were not really concerned about this question, road tunnel safety had been given consideration in many countries before 1999. In addition to experience gained by consultants, contractors and operators, research works had been conducted to develop basic and technical knowledge, mainly on tunnel fires. However only a limited number of countries had regulations in this field.

Most work aiming at producing international syntheses and recommendations was carried out by the World Road Association (PIARC: www.piarc.org). The technical scope of PIARC Technical Committee on Road Tunnel Operation created in 1957 is geometry, equipment, safety, operation and environment of road tunnels. It does not deal with the constructional aspects, which are dealt with by the International Tunnelling Association (ITA: www.ita-ailes.org). Since 1996, both associations have been co-operating on the topic of resistance to fire of tunnel structures.

Three recent reports issued by the PIARC Committee deal with: Classification of tunnels (1995); Road safety in tunnels (1995); Fire and smoke control in road tunnels (1999).

New developments since 1999

In individual countries

Immediately after the Mont Blanc tunnel fire, besides the judicial enquiry, a technical and administrative investigation was ordered by the French and Italian governments and resulted in two national reports and a joint bi-national report. 41 recommendations were made to improve the safety of this tunnel and similar ones, including information and training of users and stricter regulations concerning the size and flammability of vehicles.

In France, a check of all tunnels longer than 1 km was carried out within 3 months. A new regulation on road tunnel safety was published a year later, but could only apply to tunnels owned by the State. A law was issued in 2002 in order to apply similar procedures to all tunnels, whoever their owner. In Switzerland a tunnel task force examined the overall safety of road tunnels and made recommendations regarding the users, operation, infrastructure and vehicles. Similar steps were taken in other European countries such as Austria, Norway, etc.

At the European level

In order to harmonise the national initiatives, the Western Europe Road Directors created a working group composed of representatives of the Alpine countries and finally approved common recommendations in September 2000.

This work was resumed and enlarged by the Economic Commission for Europe of the United Nations Organisation (UN ECE: www.unece.org). Located in Geneva, this body covers 55 countries and manages a number of European agreements, e.g. in the field of road signing and road traffic, transport of dangerous goods, etc. UN ECE established a multidisciplinary group of experts on road tunnel safety. Their final report was published in December 2001 and includes recommendations on all aspects of road tunnel safety: road users, operation, infrastructure, vehicles. This report was approved by all member countries and will lead to amendments to the European agreements managed by UN ECE.

The European Union became also involved, further to a request by its Heads of States. In a first step, they included tunnel safety in their 5th framework programme for research and development. Significant research projects and thematic networks were funded, such as DARTS (www.dartsproject.net), FIT (www.etnfit.net), UPTUN (www.uptun.net), SIRTAKI (www.sirtakiproject.com), SAFE TUNNEL (www.crfproject-eu.org), Safe-T (www.safetunnel.net), etc.

In a second step, the European Commission decided to prepare a directive on minimum safety requirements for tunnels in the Trans-European Road Network. This is a legislative document, which would become compulsory in all member countries once approved and transposed into national legislation. The directive 2004/54/EC of the European Parliament and of the Council of 29 April 2004 is published now.

At the international level

Further to the 1999 fires, the PIARC Technical Committee on Road Tunnel Operation decided to lay still more emphasis on safety. Its working groups have produced the following new outputs:

- Cross-section geometry in unidirectional road tunnels (2001).
- As the conclusion of a 6-year joint research project with the Organisation for Economic Co-operation and Development (OECD: www.oecd.org), a common report was published on Transport of dangerous goods through road tunnels (2001).
- And the reports of the following Working Groups to be published soon:
 - o WG1 (Operation): Report on 'Examples of good practices for the operation and maintenance of road tunnels'
 - o WG3 (Human factors of safety): Leaflets on 'Safe driving in road tunnels', produced with the European Commission
 - o WG4 (Communication systems and geometry): Reports on 'Traffic incident management systems used in road tunnels' and 'Cross-section design for bi-directional road tunnels'
 - o WG5 (Dangerous goods): Finalisation of the Quantitative Risk Assessment and Decision Support models jointly developed with the OECD
 - o WG6 (Fire and smoke control): Report on 'Systems and equipment for fire and smoke control in road tunnels'

In the framework of the aforementioned co-operation with PIARC, ITA is finalising a report entitled 'Guidelines for structural fire resistance for road tunnels'.

Lastly a new study cycle of the PIARC Technical Committee C3.3 "Road Tunnel Operation" has just been started; it includes 5 working groups:

- o WG1-Tunnel operation and management
- o WG2-Management of tunnel safety
- o WG3-Human factors for tunnel safety
- o WG4-Detection, communication, evacuation
- o WG5-Ventilation and fire control

2.1.2 Regarding rail and metro:

A limited number of requirements specific to the safety of rail tunnels can be found in national regulations. As a matter of fact, safety is globally much higher in railway systems than on roads, and tunnels are not considered a specially dangerous part of the railway systems. Safety regulations which apply to the whole railway also improve safety in tunnels. More specifications on tunnels have been issued by the network owners.

The same still more applies to metro tunnels, and few national regulations specifically deal with their safety (only from Austria, France writing out a national technical instruction specific to metros, tramways..., Germany, Spain, and from American standards as NFPA). The situation is opposite for stations, which are generally submitted to the regulations concerning buildings open to the public. Indeed the probability for a train to stop in a tunnel and not in a station is very low, and even in such a case, the stations will normally provide the evacuation routes. A number of standards are available for the rolling stock and networks. As the characteristics of the rolling stock have a large influence on safety, especially fire safety, there are specific safety concepts for each network, if not each new line.

UITP study

In 1995, after the Bakou fire incident, UITP (Union Internationale des Transports Publics - International Union Public Transport, www.uitp.com) started a collective work on fire safety for metro. The results of this study were quite helpful for the compilation of FIT Workpackage 3 'Fire Safe Design' regarding metro.

UIC harmonisation

The Paris-based International Union of Railways (Union Internationale des Chemins de fer - International Union of Railways – UIC: www.uic.asso.fr) is the roof organisation of railways worldwide. It issues leaflets, which are considered the state of the art. In 2001-2002, a working party of 14 railway infrastructure managers and operators produced a new leaflet on tunnel safety, which was published in August 2003 as UIC-Codex 779-9. It covers new and existing tunnels over 1 km in length with mixed passenger/freight traffic of normal importance, but not very long tunnels (over 15 km). It is a compendium of over 50 measures in the fields of infrastructure, rolling stock and operations. Each measure is described in detail, considered in terms of its cost-effectiveness and gives rise to a recommendation, which distinguishes between new and existing tunnels. The UIC works furthered the compilation of FIT Workpackage 3 'Fire Safe Design' regarding rail.

UN ECE group of experts

After the finalisation of the report on road tunnel safety as mentioned above, UN ECE launched another multidisciplinary group of governmental experts to deal with rail tunnel safety. This group limited their work to heavy rail main lines, as likely to be found on international and interoperable routes. Their recommendations were finalised in December 2003. They apply to all railway tunnels, but they can be reduced for tunnels shorter than 1 km and should be adapted or enhanced for very long tunnels over 15 km. For new tunnels, the report provides an overview of best practice, similar to the UIC leaflet; in addition it proposes some measures which could become minimum standards in the 55 member states. For existing tunnels, some recommendations are given and aim at minimizing the risk of accidents.

Technical Specifications for Interoperability

The European Association for Railway Interoperability (Association Européenne pour l'Interopérabilité Ferroviaire – AEIF: www.aeif.org) has started to draft a Technical Specification for Interoperability (TSI) for Safety in Railway Tunnels. AEIF is the joint representative body mandated by the European Commission to lay down the TSIs. It brings together representatives of infrastructure managers, railway companies and industry. It has been co-founded by UIC, UNIFE (Union of the European Railway Industries: www.unife.org) and UITP and is supported by the European Commission. The relevant working group has to propose the measures to become mandatory in new and upgraded tunnels on interoperable railway lines all over the Europe.

2.1.3 Convergent safety objectives

A basic point is to define the objectives for tunnel safety, and consequently fire safety. A real convergence has appeared thanks to the international work of PIARC (report 'Fire and smoke control in road tunnels' of 1999) for road tunnels, UIC (leaflet 779-9 of 2003) for rail tunnels, and the UN ECE groups of experts for both road and rail tunnels. Some differences nevertheless exist between road and rail, due to their different characteristics and operation.

The general consensus is to give priority to the prevention of accidents and any critical events which may endanger human life, the environment and tunnel installations. This is important in all transport modes, but more efficient for rail and metro, which can achieve accident rates much lower than road. To limit accidents will also limit major fires. In road tunnels, most fires are initiated by the self-ignition of a vehicle (without any accident); however all known fires which entailed fatalities were the result of an accident, with the very important exception of the Mont Blanc tunnel fire (self-ignition of HGV).

As a second priority, the consensus is of course to limit the consequences of an accident if it has nonetheless occurred.

At this stage, road tunnels should create the prerequisite for:

- people involved in the incident to rescue themselves,
- road users to intervene immediately to prevent greater consequences,
- protecting the environment,
- limiting material damage.

Rail tunnels have the following order of priority:

- mitigate the impact of accidents,
- facilitate escape,
- facilitate rescue.

Clearly the final objective is the same: to save the people involved. In road tunnels, operating staff or rescue teams are not available on the spot in the first minutes, so that the priority is self-rescue and intervention by the users; this requires a number of measures such as detection, smoke control, emergency exits, etc. In rail tunnels, train drivers and crew are trained and available immediately; on the other hand evacuation from the train requires time. Priority measures are first to drive the train out of the tunnel as far as possible, limit the importance of the fire (including through rolling stock measures), limit the spread of smoke, and only after these, to facilitate escape and rescue in the tunnel.

2.2 General data on tunnels

2.2.1 The tunnel population in Europe

In cumulated length, Europe belongs several thousands kilometres of road, metro and railway tunnels, the latter being the major part of the whole.

Except for metro, the difficulty is to know the total cumulated length of the European tunnels, a significant part of which are very short structures ; for road and rail we will therefore indicate below, as a first indication, only the estimation of tunnels over one km.

For road

The table below [UNECE, 2001] states for every country the number of road tunnels, total tunnel length, average daily traffic and the average daily tunnel traffic (in italics, approximation for Japan). It appears that most long road tunnels are placed in countries with many mountains, like Norway, Italy and Japan. The traffic density in the tunnels vary significantly and considering the "road-tunnel-countries" based on the tunnel traffic volume in tunnels it appears that Italy, France, Switzerland, Germany, Austria and Norway is the top-six.

Country	Number of road tunnels, N >1 km	Total length of road tunnels, ΣL [km]	Average AADT [vh/day]	Tunnel traffic (AADT · ΣL) [10 ⁵ v hkm/day]
Italy	177	340		> 40
France	46	133	20620	27
Switzerland	67	162	16690	27
Germany	38	69	38670	27
Austria	55	177	11220	20
Norway	199	522	3500	18
Spain	25	58	9450	5
UK	7	13	32390	4
Croatia	9	27	5680	2
Turkey	8	17		2
Belgium	7	11		2
Russia	5	13		2
Netherlands	4	11	13000	1
Sweden	3	7	19730	1
Japan (1)	100	300		>50
USA	41	72		>15

(1) estimation for Japan

For rail

The table below shows the European countries with rail tunnels of length 1 km or longer. Eight countries have more than 100 km tunnel, Italy has most tunnels and a total length of 734 km tunnels. Germany, Switzerland, France and Austria have more than 200 km total tunnel length.

Country	Number of rail tunnels, N >1 km	Total length of rail tunnels, ΣL [km]
Italy	180	734
Germany	131	382
Switzerland	72	366
France	75	256
Austria	39	246
Norway	26	126
UK	17	114
Spain	42	110
Netherlands	6	29
Sweden	5	18
Denmark	2	12
Belgium	4	11
Greece	3	10
Portugal	3	4

For metro

Metros are different from roads and rails due to the fact that they in large majority run in tunnels. For many countries, the underground part is more than 95% of the total part. The table below presents the main data for European metros. Statistics by UITP give the number of networks, the fleet (number of wagons), the lines, the stations, the length of routes (underground and aerial) in European countries.

❖ Comments of road / rail / metro comparison

Country	Network	Fleet	Lines	Stations	Length routes [km]	
					Total	Undergr
Spain	5	2439	24	463	837	
United Kingdom	5	4274	13	390	519	
Russia	6	5669	21	261	400	
Germany	4	2107	22	387	367	
France	4	4163	24	425	304	
Norway	1	207	5	101	119	
Sweden	1	800	3	100	108	
Italy	2	1091	5	133	106	
Netherlands	1	153	5	16	73	
Romania	1	502	4	45	63	
Austria	1	257	5	86	62	
Czech Republic	1	504	3	50	50	
Belgium	1	217	3	64	41	
Hungary	1	403	3	42	31	
Portugal	1	347	4	40	27	
Finland	1	42	2	16	21	
Greece	1	168	2	19	18	
Poland	1	108	1	14	14	
Denmark	1		2	11	11	5

It appears from all indicators (networks, fleet, lines, stations and length of roads) that the 5 main metro-countries in Europe are Spain, UK, Russia, Germany and France. The indicators of these countries are exceeding those of other European countries with more than a factor 3 to 4.

2.2.2 Length of tunnels

The information below do not concern the case of short tunnels smaller than 200 m, which have the advantage – from a safety point of view – of a short distance to go out of the structure.

For road, almost all heavy traffic tunnels, i.e. of urban type, do not exceed a few kilometres, and present a daily traffic that may exceed 100,000 veh/day. Inversely, in inter-urban (country) tunnels, several tunnels exceed ten kilometres (Saint-Gothard 16.9 km; Arlberg 14.0 km; Fréjus 12.9 km; Mont-Blanc 11.6 km; Gran Sasso 10.2 km), but with a traffic amount ten or twenty times lower than the first ones. Norway has long tunnels, the longest is the Laerdal tunnel, 24.5 km long, but they show a very modest traffic.

For rail, most varying lengths are encountered, the longest are once again for inter-urban (country) tunnels. The longest operated European tunnel under operation is the Channel tunnel, operated by Eurotunnel and 50.5 km long; the longest tunnel in the world is the Seikan tunnel in Japan with 53.85 km. But somewhat longer trans-alpine structures are under project or under construction, e.g. the Mont-Ambin tunnel of Lyon Turin Ferroviaria (54 km), Brenner (55 km), or the Gotthard of Alptransit (57 km). In Europe and Japan more than 10 tunnels over 20 km is expected to open in the next 10 years.

For metro, we can consider, from the fire safety viewpoint, that an underground line is not a unique tunnel, which could then exceed easily ten kilometres, but is made of successive short tunnels separated by stations. These tunnels generally are 500-600 m long.

2.2.3 Number of tubes

Mono-tube or bi-tube configurations do exist for the three transport modes:

- for road, heavy traffic amounts generally concern bi-tube tunnels; inversely the smaller traffic densities of the longest country tunnels use mono-tube tunnels;
- for rail, the mono-tube is the most largely used, but recent long and heavily trafficked tunnels are bi-tube;
- for metro, the mono-tube is a majority, but cities like London have many bi-tubes.

2.2.4 Cross-section

As trains and metros are driven on guided tracks, the lateral spaces with respect to the trafficked section are optimised and generally smaller than for road; the resulting tunnel cross sections are often less wide and inserted recesses are often planned to protect the personnel.

Elevated and well limited walkways are very often planned for road, while this rule generally does not exist for rail or metro.

The cross-section generally is larger for road, except when ventilation ducts are installed at ceiling (ventilation of transverse type).

2.3 Tunnels in safety in view of the general operation for the three transport modes

2.3.1 Road transport

The operation of road traffic underground profits by a large number of safety arrangements aimed at correcting – with respect to open roads – the inconveniences of traffic in a restricted space. Although the probability of accidents recorded underground is smaller than in open, two essential considerations led to introduce reinforced safety measures in tunnels:

- the road transport is by far the less safe of the three transport modes
- the consequences of an accident, especially a fire in a tunnel, can be much more severe than in open, and it must be endeavoured to limit them.

Moreover, even if a few long road tunnels were built early in the twentieth century, the fire safety aspects became critical only during the latest decades, as a result of the strongly increasing traffic and stronger transportation capacities of lorries.

Therefore most European countries now have minimal safety regulations, sometimes with very detailed specifications like in France.

2.3.2 Rail transport

The railway operation in tunnels benefits from a longer experience (the first tunnels date back the years 1830 in England and France) than the road transport mode and shows a better safety level for transportation. If we dismiss the historical period where the steam traction (at a smaller degree the diesel traction nowadays) could lead to intoxication problems, passing through tunnel was not considered as a worsening risk factor, but rather like a sort of reducing factor due to cancellation of traffic hazards existing in open. For this reason, a large part of safety items in tunnel is covered generally by the rather strict rules of a railway network operation, and the safety arrangements specific to tunnels are less numerous than for road. The recent disaster fires in tunnels induced to re-examine this confident approach.

2.3.3 Metro transport

Here also operation is based on a long experience, over one century for some cities (London 1863, Glasgow 1896, Paris 1900, Berlin 1902). Inversely to road or rail, these networks are almost fully underground and the safety arrangements are especially adapted to this environment. As for rail, this mode of transport is very safe.

The safety rules are partly common with those of rail, but they are always complemented by standards regarding the design of the rolling stock and stations, especially concerning the fire risk; these standards increase the safety level in tunnel.

Resulting from the development of metros – first in large cities – a number of rules sometimes depend more from the feedback on experience and good practices than from national regulations.

2.4 Traffic nature and potential fires

2.4.1 Specific features of traffic and its management

While the road transportation occurs with independent vehicles, the railway traffic always uses convoys for a length possibly from about hundred metres (metro) to about 800 m and more (Eurotunnel shuttle, long goods trains).

For road, traffic is managed on the basis of signing facilities conventionally used on open roads, complemented with some specific devices or signs, permanently being improved and standardized on a European level. Except in case of toll station at the tunnel portal, the access of vehicles cannot be controlled individually. The equipment of road tunnels for traffic management is most varying according to the importance of the tunnel and its connection with a management centre with a permanent staff or not. The same is valid for the other safety equipment to be managed. The trend is to transfer information from tunnels to the nearest permanent traffic control centre.

For rail and metro, signing is always remote-controlled from a control centre and trains are followed individually under actual time conditions.

Speeds in tunnel for road, rail or metro generally are of about the same order of magnitude, except for high speed trains. These are characterized, from the safety viewpoint, by a short stay underground and a long stopping distance: this is favourable regarding safety since the probability to stop within the tunnel is low.

2.4.2 Characteristic features of vehicle driving

The road transport mode requires by definition that the driver keeps permanently and voluntarily the vehicle on its lane.

The heel of Achilles of safety in road tunnels is the adequate behaviour of the great number of drivers passing through, a mixture of occasional and regular or professional drivers, of young and very old drivers... In spite of the prevention actions of the authorities, parameters that can hardly be controlled as consuming alcohol, narcotics or medicines, or simply the effect of tiredness or stress increase the risks of accident. According to statistics, however, the accident rate in tunnel is lower than that on open roads.

Based on European statistics it is estimated that fire occurs with a frequency of approximately 4 - 5 fires per 100 million vehicle km. Less than 1 % of the fires will be characterised as fires with the serious consequences (fires involving injuries, fatalities or large material damage) but these fires have mostly the result of an accident. The Mont Blanc tunnel fire, which was caused by self-ignition of a heavy goods vehicle, is here an exception to the rule. The main causes of fire in tunnels, referring to PIARC are shown in the table below.

❖ Comments of road / rail / metro comparison

Causes of fires	Distribution
Accidents	20%
Electrical, mechanical and other reasons	80%

Cause of serious fires	Distribution
Motor and gearbox	45%
Collision	36%
Brakes and wheel	15%
Lost items	3%

Railway and metro are rail-guided transportation means, for which the risk of route deviation is highly improbable and reduced to that of derailment or wrong shunting. Drivers are all professionals, permanently trained, with the possibility to make them observe safety guidelines specific to tunnel crossing.

2.4.3 Transported people

For road, the vehicles usually transport only one or several people, but there is always a part of the public transportation (mini-buses and buses) which can concentrate from 8 to about 60 people and more on the same vehicle.

While a goods train hosts only one or two people, a passenger train transports several hundred people, even more than thousand people (from 100 up to the extreme number of 250 per coach).

For metro, as for trains, the metro trains can transport several hundreds people (100 to 150 per coach).

2.4.4 Potential fires

Data on fire in tunnels are provided for the three transport modes on the mentioned FIT website and the FIT Technical Report – Part 1 'Design Fires'.

For road

The road vehicles are all driven by internal combustion engines and include gasoline or gas-oil tanks (several tens litres for passenger cars and up to more than 1000 litres for some international transportation lorries). At present the liquid gas driven vehicles are in a minority. Every vehicle, of a more or less sophisticated technology, integrates in itself all ingredients that may lead to a fire: hot parts of the engine auxiliaries, brakes, fuel reserve, circulation and injection of fuel, numerous electrical circuits, more and more important quantities of plastic material and rubber... Regarding the HGVs, in view to improve the performances of the engines and reduce the emitted pollutants, the automobile factories also design turbo-compressors and exhaust silencers operated at a higher temperature than in the past. Concerning the buses, however, standards have been set up on the fire behaviour of materials used for the inside equipment (disaster fire of a bus in open in Beaune, France).

It should be noticed that the manufacturers do not integrate at all the objective of a reduction of fire risk in tunnel in the design of vehicles, the flammability of which is high.

Tunnels are sometimes reserved for only one category of vehicles, like passenger cars in reduced size tunnels, but most tunnels are passed through by a composite traffic of passenger cars and lorries. Except for lorries transporting dangerous goods, for which the access to the tunnels is strictly controlled (prohibited or authorized, but often under certain conditions), the access of vehicles is free. Concerning the lorries, this free access opens to a large variety of caloric potential of the loading, from non-inflammable or lowly inflammable (minerals, metals, plants...) up to highly inflammable (wood, plastic materials, grease...). Such loading – which can represent several tens of tons – is unknown from the operator at the fire time.

The heat release rate of a burning vehicle may be from 2 to 100 even 200 MW.

While the fire source may develop as well in a passenger car and in a lorry, the inflammation of a lorry obviously is the major risk in a tunnel, and can lead to a disaster.

For train

As a general rule the traction technology is located at the end of the convoy, using electric and sometimes diesel motor coach. The fire risk is concentrated rather on these machines, with for diesel a risk component related to the presence of gas-oil, but a fire can start on wagons (hot boxes...).

As for road, some tunnels can be reserved to only one type of transportation, for instance passengers in urban undergrounds or very high speed country undergrounds, but the composite tunnels – passengers and goods – are the most numerous, they often are the most worrying considering fire safety. Operating measures can allow the passage of only one train in the tunnel at the same time (e.g. dangerous goods), but this has an impact on the line capacity.

The design of modern passenger cars with respect to the fire behaviour of the materials meets certain standards; these are sometimes still stricter in some countries when the trains are aimed to be operated underground, and therefore are operated somewhat like a metro.

Regarding the goods trains, like for road, there is an infinite range of possible loadings, also with regulations for the dangerous goods. The caloric potential loaded on each coach is close to that of lorries, knowing that this can be lorries themselves or passenger cars transported by shuttles. In this latter case the risk of fire to lorries is not so high as for road, because they are no more running and their condition can be checked before the train departure (fuel loss, hot points).

The heat release rate of a train fire may be from 20-25 MW (Diesel locomotive, passenger carriage) to about 50, 100 and even 200 MW (open freight wagon with lorries).

With respect to road the immediate proximity of successive wagons strengthens the problems of fire transmission between the units.

The load transported by a train is ten to over fifty times higher than that of a lorry.

❖ Comments of road / rail / metro comparison

For metro

The trains are driven exclusively by electric traction, and – inversely to the road vehicles – built in view of operation within tunnels. The fire risk is minimised on the recent equipment especially thanks to the regulations regarding the fire behaviour of materials.

The heat release rate of a metro fire may be from 6 to about 25 MW.

2.5 Action towards fires

2.5.1 Vehicle on-board means for fire detection and fighting

No strictly speaking smoke or fire detector is available on road vehicles, only sensors providing information on the operating conditions of the vehicle.

For rail, according to the age and nature of trains and metros, alarms can be planned for technical anomalies such as axle heating, derailment or fire detection in the engine coach. The tourist and HGV shuttles of the Channel tunnel are fitted with fire detectors in every wagon.

In the passenger coaches or metros, a starting fire can be reported to the driver, like any other danger, as soon as a passenger activates the alarm signal handle. But this signal means a severe safety problem in tunnel, because it causes braking and emergency stopping of the train. The control of air conditioning can be also a safety problem.

The extinction means planned on board generally are limited to portable extinguishers, planned systematically for rail and more indefinitely for road. Fixed on-board extinction systems – or rather mitigation systems – already exist on some locos, as well as in certain types of trains like the Eurotunnel tourist shuttles (halons) and soon HGV trains (water spray), also in the Madrid metro coaches, but such cases are exceptional.

2.5.2 Fixed means for fire detection

In road tunnels various equipment can be used to alarm the operator: camera surveying, specific fire detectors in some tunnels (heat or smoke), pollution sensors, safety door opening alarm, etc.

In railway tunnels there generally is no detection system in the interior zone. The Channel tunnel is an exception.

In metro tunnels, detectors are available in stations, technical rooms or commercial premises.

2.5.3 Exchange of information with the users

For road, to allow a distressed user to exchange information with the surveying operator, he must have an access by foot to the phones distributed all along the tunnel or within protected recesses. As a general rule channels of cellular phones are not re-transmitted inside tunnels.

From the control centre, the operator has no possibility to communicate with the driving users. In tunnels with the best equipment he only can send visual information via varying message signs or information audible in the vehicles on public radio channels. Sometimes – this is less frequent or less efficient due to the reverberant sound – loudspeakers installed within the tunnel can be used.

For rail and metro, a ground-train radio-connection between the operator in the control centre and the driver is possible under normal conditions.

In the passenger trains and metros, coaches are wired for sound, thus allowing the train crew to broadcast messages audible to all the passengers. In the metro stations or underground railway stations, a loudspeaker relay allows the head of station to inform the people evacuated from the train about the adequate behaviour.

Fixed emergency phones for the users are available on the platform of metro stations and service phones are generally planned in the metro and railway tunnels.

2.5.4 Ventilation and smoke control in case of fire

The range of ventilation modes is quite larger for road than for rail or metro.

Road tunnels

Attention has been given for a long time to the sanitary ventilation of road tunnels, firstly due to the problems of dilution of high pollutant quantities emitted by the vehicles. The importance of smoke control has been recognized only during the latest twenty or thirty years.

Any road tunnel of significant length is equipped with an artificial ventilation. This may be:

- either of longitudinal type, the most simple and economical system, allowing to push smoke along the tunnel in the desired direction in case of fire;
- or of transverse type, a more expensive system, however allowing to extract smoke at the ceiling at any point of the tunnel to prevent longitudinal extension along the whole tunnel section.

Railway tunnels

A mechanical ventilation there is rare because the electric traction is the most frequently used nowadays, the piston effect of trains is high, and the intermediate ventilation shafts allow proper sanitary conditions in most tunnels.

A few tunnels only are equipped with a longitudinal ventilation, principally to control smoke in case of fire; the transverse system is never used.

Metro

Many lines are fitted with ventilation for comfort and fire smoke control. All facilities are planned on the basis of the longitudinal scavenging of tubes, with various blowing/extraction models in the stations or by shaft in the central part of tubes.

2.5.5 Fixed means for fire fighting

Portable extinguishers and hydrants - sometimes with water-hose nozzles – are generally distributed at regular interval along the road tunnels. Metros are equipped in a similar way in each station. This equipment is scarcely available in the interior zone of the railway tunnels. The fixed water spraying systems in tunnel are not developed in Europe, except one or two cases, but they are under study.

2.5.6 Escape of users

Metros and some urban road tunnels (cut-and-covers) are located at shallow depth, thus facilitating the creation of staircases to the ground surface.

In mono-tube for railway, there is generally no other exit or access than the tunnel itself; for mono-tube for road, and according to the countries, there are solutions of shelters (in France) or ways independent from the traffic space and accessible to the pedestrians (parallel gallery, ventilation duct or direct communication to the surface). For metro, the stations ensure the pedestrian communication to outside via staircases or escalators; this can be the case for underground railway stations too.

In bi-tube, inter-tube communication generally exists both for road and rail, but the space is more restricted for road, about 200-400 m instead of 600-800 m and more.

The conditions to evacuate by foot the passengers from a train or a metro in full track – i.e. between two stations - are more difficult (if not impossible) than for road vehicles, due to:

- the great number of people to be evacuated, which amplifies the phenomena of panic and obstruction of emergency exits
- the absence of platform and the height of the coach floor about one metre above the track
- of the often restricted passage width between the train coaches and the tunnel sidewall, and due to the difficult walking on the ballast when this is possible, for instance in a two-way mono-tube.
- for metro the risk of electrocution by live rails that must be cut first.

Moreover the trains have communicating doors between the coaches, but this escape way through the train lays a problem of quick saturation by the evacuated crowd and of a phenomenon of panic.

The logic of the emergency escape from the metro lays on the necessity for the driver to reach a station platform and from the train on the possibility to reach the open. Especially for railway there is a risk if the train stops to destroy the power supply of the locos (the overhead line is most exposed to fire) and hinder possibilities to re-start.

The safety lighting within the tunnel and emergency exits is one of the major safety measures for all three transport modes. Under normal operating conditions, the road tunnels profit by a high pavement illuminance level required for traffic safety. For their part, metros profit by the good lighting of platforms and accesses.

The presence of cameras in major road tunnels and in metro stations allows a best assessment of the escape conditions of people in the non-smoky areas; this is not the case for the interior zone of railway tunnels.

2.5.7 Intervention of rescue services

Concerning the intervention logistics of rescue services, the traditional vehicles can always use the road tunnel lanes when free, while the presence of rails and ballast in a railway tunnel means complicated manoeuvres. Emergency vehicles on rail or composite rail/road are not used currently. Examples of platforms that road vehicles can access to in railway tunnels do not seem to be available.

The intervention time of firemen in tunnel is optimised for metro (possibility of 5-10 min only) thanks to the urban environmental conditions and since they can access from the station. The intervention time for road can also be short if relevant staffs are available at the portals, but it can reach, like for rail, about half an hour to one hour in the other cases, thus representing a rather long time sufficient to have the fire cause human damages.

For the three transport modes, it therefore appears that, due to the intervention time of rescue services, the quick self-escape of the users is the prime priority in case of fire.

2.6 Comparative synthesis table

The comparative table below, inspired from a document of IUPT (International Association of Public Transport) gives a typology of the main safety elements in tunnel for the three transport modes.

It appears that the approach of the safety level and its improvement for each mode corresponds to a certain diverging problematic and to specific technical cultures.

Due to the higher risk level in road tunnels than in railway or metro tunnels, road required to define more important safety measures and to write out more developed regulations and guidelines than for the other tunnels.

But the potential fire does not know which type of tunnel it will start in; this is the reason why recommendations to limit its consequences should be established according to the most pertinent and unified assessment methods. This certainly is one of the major objectives of those studying now this topic on a national, also European and even international level.

❖ Comments of road / rail / metro comparison

Item	Metro	Rail	Road
Length	5 to 600 meters mean between 2 stations	30 m to about 50km	200 m to about 20 km
Location	city	city, country	city, country
Exits	stations	tunnel ends	tunnel ends, shelters with access to other tunnels
Possibilities to move from accident place to safe exit	very narrow pathways	narrow pathways	wider pathways
Intervention time of firemen	5 to 10minutes	10 to 60 minutes	5 to 10 (firemen at the end) to 60minutes
Fire heat release rate	7 to 20 MW fire load under control	10 to 200 MW(TMD) fire load depends on vehicles (their load)	2 to 200 MW(TMD) fire load depends on vehicles (their load)
People	100 to 250 per wagon	150 per wagon	1 to 100(bus)
Traffic control	strict control	strict control	no control to individual drivers
Communication for alarm	driver or interphone	driver of the train	each driver of each vehicle
Materials	fire resistance standard	fire resistance standard	no standard
Firemen intervention	stations cannot use cars	ends of tunnel cannot use cars	ends of tunnel, special accesses

Chapter 3 : Conclusions on the Compilation of guidelines for fire safe design for road, rail and metro tunnels

3.1 Main features identified by the guideline compilation

After achievement of the works conducted on the compilation of guidelines for fire safe design the following conclusion can be given.

The existing texts of national regulations regarding the safety arrangements for tunnels are largely more numerous for road tunnels, which present higher risks intrinsically, than for the rail or metro tunnels.

The dramatic fires which occurred in the road tunnels of Mont Blanc (France-Italy; 39 fatalities) and Tauern (Austria; 12 fatalities) in 1999 have caused a radical change of views on tunnel safety. This topic, which was previously reserved for specialists, became a real concern for the European public opinions, which triggered politicians to be involved. This concern was reinforced two years later by the fire in the Gotthard tunnel (Switzerland; 11 fatalities). Rail tunnels were also affected by fire catastrophes, such as in the Channel tunnel (UK-France; no fatality but very severe damage) in 1996, Kaprun funicular tunnel (Austria; 155 fatalities) in 2000 or Daegu metro (South Korea; 200 fatalities) in 2003.

The logical answer of the authorities to these events was to launch the drafting of new rules based upon an exhaustive re-examination of the fire safety problems. This examination aimed at improving safety in road tunnels finally also integrated the problematics specific to rail and metro - which were so far deemed as much safer but which concentrate a high number of users – and rapidly reached the European legislative framework. The working groups of the international organisations scheduled for their part a great number of new actions on this issue.

Regarding the assessment of the role, efficiency and adequacy of the technical safety measures, it can be stated that – especially for the major tunnels – there is a clear tendency of the recommendations to advocate risk or hazard studies based on design fire scenarios, in order to validate the consistency and the proper level of the whole safety system.

Regarding efficiency it may be useful to strive to play on the equivalence of measures of various nature, for instance in view to reach a comparable safety level at the lowest cost. But the definition of a scale in the quantified assessment of the cost-effectiveness remains a difficult task. The imperfection of the analysis essentially comes from the rare feedback on experience of very severe incidents and from the obviously quite simplified hypotheses selected with regard to the great number of concerned parameters. Controversy may appear regarding the modelling of the human behaviour, still insufficiently known, or regarding the needs of translation into cost; not only of the economic loss related to the interrupted tunnel operation and repair works but principally of the loss of human lives.

In the definition of the means necessary to fulfil a given safety function, the fire engineering approach based on design fire scenarios is a more and more useful and promising study complement, for instance to evaluate the behaviour of a structure or equipment, and possibly adapt it to the requirements. But the essence of the examined existing guidelines, however, consists in prescriptive (or performance based) elements that delete the problem of a possible variation in the definition results of safety measures according to the hypotheses or

techniques and computation means used by the designer. The prescriptive approach often allows – at least partly – to refer to the same standards as already widely used in other fields, e.g. in building trade or industry (for instance temperature-time curves for fire resistance tests). The prescriptive approach has the advantage to give a more simple and universal definition of the minimal safety arrangements, and on the spot it also allows to get a certain harmonization between the structures of the safety arrangements as perceived by the user or used by the emergency services.

The stake of safety in tunnels induces the designers and builders to search for numerous innovating techniques, but this aspect generally is not directly apparent in the guidelines formulation.

3.2 More specifically for road tunnels

The compilation report for road includes a detailed comparison which presents the requirements of the national guidelines of Germany, France, UK, Norway, Austria, Switzerland and Netherlands, to which we added the requirements of the new European directive, which is the first community regulation on this topic.

The substantial ideas that can be deducted from the compilation are the following:

- The notion of traffic and underground length is determining in the definition of the safety measures; this allows several countries to define tunnel categories (UK, Austria, Norway, France). The presence of lorries transporting dangerous materials leads to complementary specifications.
- The emergency passenger exits to safety and the emergency access for rescue staff generally are dealt with by national regulations, precise but not homogeneous between the various countries. It can be found that inter-distances are varying from 100 m to 400 m between the escape routes; the European directive defines a maximum at 500 m if any. The requirement for shelters is not frequent and these must have an access way connected to the outside (France, European directive).
- The drainage of flammable liquid is a safety element rather well defined by certain countries, with civil engineering and geometry arrangements specially adapted.
- Among the safety equipments ventilation and smoke control in case of fire are considered as primordial and lead in most countries to detailed guidelines. The following can be compulsory: necessity of an artificial ventilation, the ventilation system, the required air volumes and velocities, or simply the objectives that must be met according to the selected design fire (performance base approach). Requirements are stated to prevent smoke penetrating into the emergency exits and rescue access.
- The lighting of the tunnel and emergency exits and rescue access is – except special cases – defined by a minimal assisted illuminance level.
- The requirements for traffic signage, both outside and within the tunnel, and signage for pedestrian exit and rescue generally are well stated in the guidelines, but criteria remain heterogeneous.

- Regarding communication and alarm systems, the emergency telephones and the alarm push-buttons generally are imposed as minimal basic elements; the required inter-spaces however are most varying: from 50 to 250m; the value of 150m as stated by the European directive therefore is a good compromise. But requirements also exist which are well focused on the automatic alarms on equipments, automatic incident detection, fire or smoke detection and on radio rebroadcast. The installation of loudspeakers within the tunnel itself is not frequent, but requested in the evacuation facilities or shelters for the users.
- For traffic regulation and monitoring equipments we notice that the measures must be adapted to the surveillance level of the tunnel. We establish mainly guidelines which allow the quick detection of the traffic incidents, such as traffic speed and density measurement or a video control, and guidelines regarding means for a quick closure of the tunnel. The thermographic portal detectors to detect the abnormally hot lorries before they enter the tunnel are never prescribed.
- The requirements for emergency power supply of the safety equipments are generally well described.
- Regarding fire fighting, the distribution within the tunnel of extinguishers and the presence of a water network and fire hydrants of sufficient capacity are a compulsory customary basis, but with varying characteristics and inter-distances. Several countries define a hydrant inter-distance between 150 and 250m, but all guidelines do not observe the maximal value of 500m stated for all tunnels by the European directive. The installation of a fixed fire suppression system is not imposed in any regulation.
- The structure and equipment response to fire are dealt with in a rather large description of the requirements, however without homogeneity. Regarding the resistance of structures, the formulation varies from very prescriptive requirements (Germany) to more or less performance based criteria (France, Austria, Norway). The criteria are given in terms of duration and specified fire curve or heat release rate. Calculated documentation is required in all guidelines. Concerning the equipments the notion of continuity of service for the safety elements is often emphasized and connected to most varying criteria of heat reaction or resistance. The European directive defines much less precise these requirements than certain national guidelines.

3.3 More specifically for rail tunnels:

Although the guidelines regarding safety in rail tunnels are rather rare and dissimilar, the compilation could be based on over thirty documents. But the references selected for a detailed comparison come from France, Germany, Italy, Spain et Switzerland.

The improvement of fire safety and the evolution of concepts were highly enhanced on the occasion of the largest projects: Eurotunnel, CTRL, Lyon Turin Ferroviaire or Alptransit. But the direct application of these concepts on upgrade activities of the huge number of existing tunnels would not be thinkable, because the context and the problematics are very different and serious problems of feasibility often appear, requiring another approach. Several countries require specific risk studies (risk based evaluation) to adjust the choice of certain measures.

For long or very long tunnels with mixed traffic (passengers and goods) it is recommended to build two tubes (Germany: category 2 and 3). The safety rules analysed in the compilation are logically clearly oriented to the safety of passenger trains of the mainlines.

The following major items can be drawn from the guidelines:

- Emergency passenger exit for users: according to the country, guidelines that are sometimes rather detailed, are given on geometry or spacing of the escape routes, but without homogeneity. The specifications for the mono-tubes are much more imprecise than for the bi-tubes which provide inter-communications. Regarding these bi-tubes, inter-distance values of 250m (Italy) and 800m (France) are given. The closed shelters without exit to open air are not permitted.
- Emergency access for rescue staff: Geometric criteria for the passage sections, permissible maximal gradients or characteristics of the access shafts are given. A special emphasis is given in Switzerland and Germany on the necessary access of road vehicles.
- Except in France for dangerous goods and in Switzerland, there is practically no rule imposed about the drainage of flammable liquids.
- Except for certain types of tunnels (France, Spain), smoke control in case of fire generally is not dealt with by precise guidelines. Specific studies are sometimes suggested (Switzerland).
- As a general rule minimal requirements on emergency lighting in tunnel and emergency exits are defined.
- Signage: For traffic no complementary measures to those applied for the operation of the open road network are given for the tunnels. Concerning the pedestrian exits and rescue accesses Germany and Italy provide rules.
- Communication and alarm system: Telephones are often requested at the tunnel portals and in the vicinity of the escape routes and special requirements are sometimes specified for radio rebroadcast for the emergency services. There is no guideline about the fire detection within the tunnel, but sometimes for the technical rooms.
- There is no tunnel specificity for the traffic regulation and monitoring equipment.
- Regarding power supply, there are some guidelines on redundancy, emergency power supply by batteries or the possibility to switch off the electrical supply for trains.

- Fire fighting: most countries require to plan a water supply with dry or filled pipes. The available water capacity and the distribution of nozzles are quite heterogeneous. There is no regulation on the fixed fire suppression mitigation system.
- Structure and equipment response to fire: No guideline for the structure except regarding the use of flammable materials (France, Germany, Italy). Regarding the equipments all countries provide minimal requirements on the resistance and operating time of the emergency systems (cables, fans).

3.4 More specifically for metros

With respect to the mainline railway tunnels the compilation of guidelines revealed a marked specificity of the fire safety aspects of the metro, although this type of transport also uses rail.

The basic differences come from the existence of stations at a small inter-distance, which play a major role for safety, and of a rolling stock specially dedicated to the transport of people and designed to limit the risk of fire underground. An additional safety principle is based on two very important points: braking inhibition in case of emergency (however seemingly not yet in general use) and the principle the get the train to a safe zone, generally the next station.

Another particular aspect is that there are only few national guidelines specific to metro. In fact there are only a few European standards (Austria, France, Germany, Italy, Spain), and the NFPA U.S. standards, which have a strong international influence to American, Asian and even European continents. Inversely detailed specifications are decided for the network of each city individually. Because of the scarcity of national standards and like the valuable comparative study on safety conducted by UITP, the FIT report 'Fire Safe Design – metro' makes reference to cities (17 European cities, plus Moscow) and not to countries. Hence the projects for new metro networks are an opportunity to complement or improve the concept of safety and apply some innovations.

Although the objective of the compilation was focused initially onto the safety measures in the tunnel part of the metro, it finally appeared as unavoidable to also consider the measures planned for the stations, which bring a direct contribution to the whole safety. The report therefore was structured on the basis of this inseparable tunnels-stations couple, contrary to the preceding two reports. For the tunnel part, criteria are seen from rail tunnels guidelines (France, Austria) or specific to the guided transports (Italy, USA, France under preparation). Criteria for the station part are rather based on rules drafted for building trade or public premises. The latter continues to evolve.

The major elements drawn from the compilation can be summarized as follows:

- the emergency passenger exit to safety and the emergency access for rescue are by nature ensured at two points at least of each station; the inter-distance between the stations is 600 m in average. The passage width is defined according to the time necessary for the evacuation by foot at peak hours. Intermediate accesses for the firemen can be added in the longest underground parts. The evacuation of a train blocked within the tunnel is generally not impossible but presents hardly solved difficulties.
- the ventilation and principally smoke control in tubes and stations are considered as primordial. The basis of the smoke control design are of descriptive order, i.e. performance based criteria.

- Normal and emergency lighting are available in stations and most of the tunnels in case of fire or power supply failure. The illuminance levels differ according to the network
- The signage consists only in signs of escape direction and distance to the station
- The alarms can be given, in stations by passengers or operator, in the train by passenger or driver (in automatic systems by cameras in the wagon)
- The communication system is coherent, multiple and complete. Information to passengers has a good efficiency due to the use in normal operation (by the driver in the train or automatic supervision, by the operator in station)
- Fire fighting: essentially for firemen by means of dry or wet pipes in some networks (generally part of them) and extinguishers for operators and drivers
- Traffic regulation and monitoring equipment are required in normal operation, and only difficult at the peak hours when the need for stopping the train in station is urgent.
- The response to fire in station is like that in buildings, but the main differences concern the cables laying in tunnels for long distances according to different functions: power, communication, control and command, safety.

3.5 Future work on fire safe design

The unprecedented disaster fires which occurred in tunnels showed that these can concern the three types of massive passenger transport selected by FIT, i.e. road, rail and metro, and that a new examination of the safety problems was required. This resulted in a high intensification of studies, initiated by various national, European or international bodies.

The work conducted by the FIT Workpackage 3 'Compilation of guidelines for fire safe design' led to the conclusion that the regulations, which are the fundamental basis of the fire safe design in the various European countries obviously often needs to be improved. It can be recognized that the national documents have a quite varying content, that they can lead to rather different safety levels for the same category of tunnels, while they do not always impose the minimal safety measures that the recent committees of experts and managers deemed as necessary.

Among the major issues that are identified for improvement, we can mention:

- the extension of exchange of experience and competency between the European countries beyond the present joint projects; this would allow to improve the safety optimisation with a more informed and more harmonious formulation of the new reference texts. The example of the new European directive on safety in road tunnels – which has still to be applied in each country – follows this tendency and could be observed for rail and metro.
- the better quantified consideration of the inter-activity of all systems that interact in a tunnel
- the complement and improvement that can be brought to the fire safety design of the performance-based approach with respect to the present and simple prescriptive approach
- the more systematic recourse to the risk studies based on experienced methods, of an adequate level and if possible standardised

- the better quantified integration of the cost/efficiency couple of the safety measures within the hierarchy of the possible choices. An aspect difficult to assess, however specially important to upgrade safety in the existing tunnels according to a “reasonable” budget to be planned
- a better identification of the human behaviour allowing to conceive more efficient safe keeping means
- the research of standardisation among the tunnels of the safety measures and of their management, enabling the users, the rescue services and the operators to understand them, memorize them and operate them more surely in case of incident
- the adequacy of the surveillance level to the type of tunnel
- the consideration of technical innovations which allow to meet safety objectives more ambitious than before
- lastly, the control procedures of the tunnel safety level, tests, exercises, education, training, and organisation of the operators and rescues services.